

Lecture 7 — Type of Memory

Memory is a functional component of a computer that stores data, instructions, and results.

Now let's understand how memory is classified.

One important distinction: memory and storage are closely related, but they are not always the same thing in computer terminology.

We'll start with the main classification.

Computer Memory

Primary Memory

Directly accessible during processing

Secondary Storage

Long-term data storage

RAM, ROM

Cache and registers are also part of the broader memory hierarchy.

SSD, HDD, USB

Persistent storage devices

1. Primary Memory

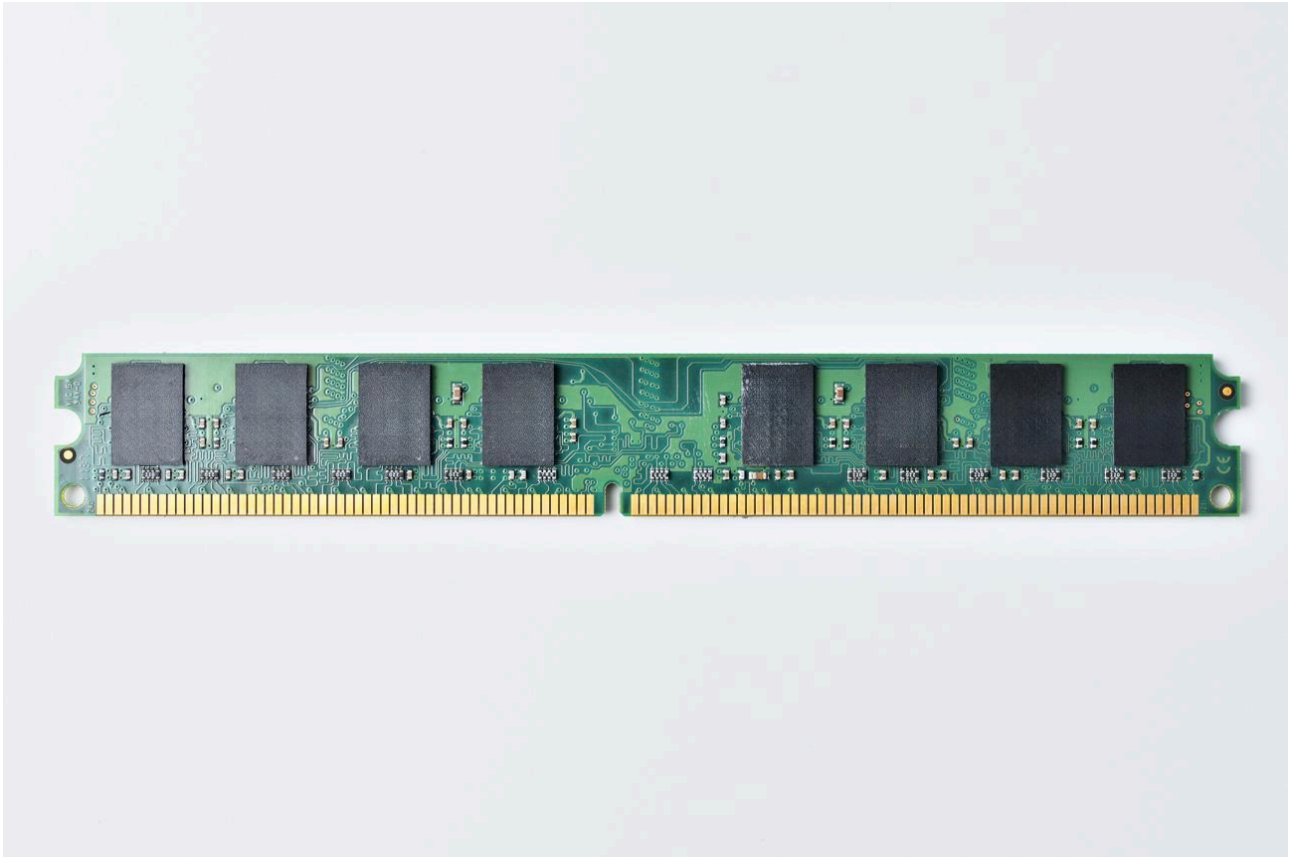
Definition:

Primary memory is the memory directly accessible by the CPU for its immediate processing needs.

It holds data and instructions needed during program execution.

Let's understand its important types.

A. RAM — Random Access Memory



RAM is temporary working memory used by running programs.

For example, when you open a web browser, its currently needed data and instructions are loaded into RAM.

Important characteristics:

- Volatile: its contents are normally lost when power is removed.
- Read and write: data can be read from and written to it.
- Fast access: it supports rapid access to data during program execution.
- Temporary: it is generally not used for permanent storage.

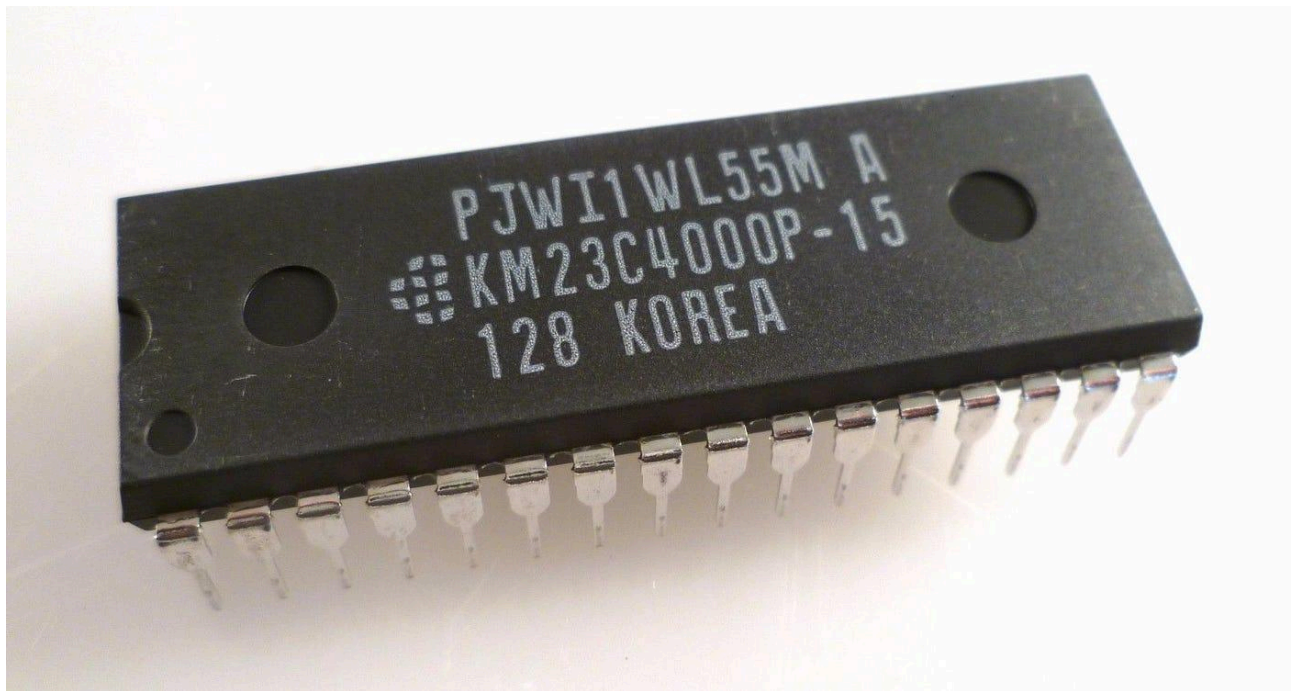
Example:

Imagine your computer has 16 GB RAM.

When you open a browser, a code editor, and several other applications, they use some of this RAM to work.

The more applications and data you keep active, the more RAM may be required.

B. ROM — Read-Only Memory



ROM stands for Read-Only Memory.

It is non-volatile memory, meaning it retains its contents even when the power is switched off.

It is used to store firmware and other essential instructions in many computer systems.

Important characteristics:

- Non-volatile.
- Retains data without power.
- Generally used to store firmware.
- Not ordinarily changed as frequently as RAM.

Modern ROM-based technologies include programmable and rewritable forms, so the name "read-only" is a simplification of how the broader family works.

RAM vs ROM

RAM	ROM
Usually volatile	Non-volatile
Used as working memory	Commonly stores firmware
Frequently read and written	Usually updated less frequently
Contents normally lost without power	Contents retained without power

2. Secondary Storage

Secondary storage is used to store data and programs for long-term use.

Unlike RAM, it generally retains information when the computer is turned off.

Examples:



SSD

Solid State Drive. Uses flash memory and has no moving mechanical parts.



HDD

Hard Disk Drive. Uses magnetic disks and mechanical components.



USB Flash Drive

Portable storage that uses flash memory.



Optical Disc

Examples include CDs, DVDs, and Blu-ray discs.

Why do we need secondary storage?

Imagine your laptop has 16 GB RAM and a 512 GB SSD.

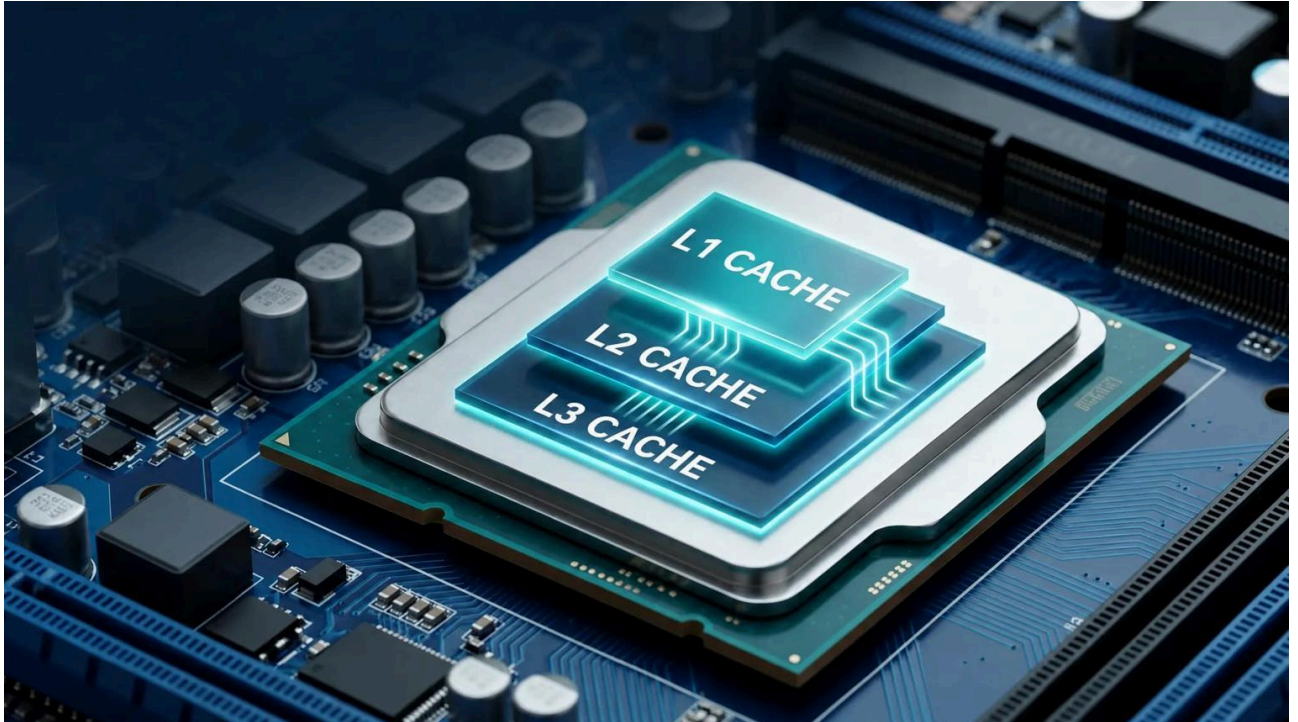
You can use RAM for currently running programs, while your SSD stores:

- Your operating system
- Python projects
- Photos and videos
- Documents

- Installed software

If you switch off your laptop, the SSD retains those files, while the contents of ordinary volatile RAM are lost.

3. Cache Memory



Cache is a small, high-speed memory that stores frequently or recently needed data and instructions to reduce the time the CPU spends waiting for information.

For example, if the CPU repeatedly needs particular data, cache may hold a copy so that the CPU can access it more quickly than retrieving it from RAM.

Common cache levels:

- L1: Usually the smallest and fastest CPU cache.
- L2: Usually larger than L1.
- L3: Often larger and may be shared among multiple CPU cores.

The exact design varies by processor.

Think of cache as a fast-access workspace between the CPU's execution units and main memory.

4. Registers

Registers are extremely fast storage locations within the CPU.

We already studied five important registers:

Register	Purpose
PC	Holds the address of the next instruction
MAR	Holds a memory address being accessed
MDR	Holds data or instructions transferred to/from memory
CIR	Holds the current instruction
AC	Holds intermediate or result values in a simplified CPU model

Registers are not usually treated as a type of secondary storage. They form the fastest, smallest level of the memory hierarchy.

5. Memory Hierarchy

Now connect all these types.

Memory Hierarchy

Simplified illustration

Registers

Fastest, smallest

L1 / L2 / L3 Cache

Very fast, small

RAM

Main working memory

SSD / HDD

Larger, persistent storage

Moving downward generally means increasing capacity and access time.

This hierarchy reflects a fundamental trade-off:

- Faster memory is generally more expensive per bit and has lower capacity.
- Slower storage is generally cheaper per bit and offers greater capacity.

6. Important Differences to Understand

Feature	RAM	Cache	Registers	SSD
Main purpose	Working memory	Faster access to frequently needed data	Immediate CPU working values	Long-term storage
Typical location	Memory modules / integrated memory	On or very close to CPU	Inside CPU	Storage device
Volatile	Yes	Yes	Yes	No
Relative speed	Fast	Faster than RAM	Extremely fast	Slower than RAM
Capacity	GBs	KBs to tens of MBs, depending on design	Very small	Hundreds of GBs or more

These are general comparisons; actual performance and capacity depend on the computer's architecture.

7. Real-Life Example: Opening a Python Program

Let's connect memory types to something you already know.

Suppose you open VS Code and run:

```
a = 10
b = 20
print(a + b)
```

Here's a simplified explanation of what happens:

1. SSD: Your VS Code installation and Python files are stored on your SSD.
2. RAM: The operating system loads the necessary program data and instructions into RAM.
3. Cache: Frequently needed instructions and data may be kept in CPU cache.
4. Registers: The CPU uses registers to hold immediate values, addresses, and intermediate results.
5. CPU: The processor executes instructions and performs the required operations.
6. Output: The result `30` is displayed in the terminal.

Notice how all the components cooperate.

This is the connection between memory and instruction execution, which will help you understand our next topic.

Exam-ready summary

- Primary memory: Memory directly accessible for immediate processing, such as RAM and ROM.
- RAM: Volatile read/write working memory.
- ROM: Non-volatile memory commonly used for firmware.
- Secondary storage: Persistent storage such as SSDs and HDDs.
- Cache: High-speed memory that reduces average access time for frequently needed data and instructions.
- Registers: Tiny, very fast storage locations inside the CPU.